Leaf Disease Identification and Remedy Recommendation System

(AN AI-POWERED OFFLINE-CAPABLE PLATFORM FOR ENHANCED AGRICULTURAL MANAGEMENT)

Leaf Disease Analyzer

Project Final Report

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DECLARATION

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ABSTRACT

Agriculture is the backbone of Sri Lanka's rural economy, where the cultivation of fruit trees, plantation crops, and vegetables provides both food security and household income. However, these crops are continuously threatened by foliar diseases such as Anthracnose, Red Rust, and Leaf Miner infestations, which silently reduce photosynthetic area, weaken plants, and cause significant yield loss before farmers are even aware of the damage. Most rural farmers still depend on visual inspection and the experience of fellow growers to identify diseases. These traditional practices are slow, often inaccurate, and heavily dependent on subjective judgement. As a result, treatments are frequently delayed, and the misuse of broad-spectrum pesticides damages soil health, contaminates water bodies, and increases production cost.

This research proposes a Leaf Disease Identification and Remedy Recommendation System that combines Convolutional Neural Networks (CNN) and Image Processing to support farmers in field conditions. A CNN-based transfer learning model, built on the lightweight MobileNetV2 backbone, is used to classify leaf images into three target disease classes — Anthracnose, Red Rust, and Leaf Miner. Each class is supported by approximately 1,200 labelled images, totalling around 3,600 training samples. Image preprocessing techniques such as resizing, denoising, and HSV-based segmentation are applied to improve image quality before classification. Once a disease is identified, the system queries a local remedy knowledge base and returns a structured set of cultural, organic, and chemical interventions appropriate for the predicted disease.

A key feature of this component is its offline capability. The trained model is converted to TensorFlow Lite and packaged inside the mobile application along with a local SQLite remedy database, allowing farmers in low-connectivity rural areas to obtain disease diagnosis and remedy guidance without any internet connection. The backend is implemented in Python using TensorFlow/Keras for deep learning and OpenCV for image processing, while the mobile application is built using Flutter for cross-platform Android and iOS support. Firebase is used optionally for synchronizing historical diagnosis records when connectivity is available. Experimental testing achieved more than 92% accuracy in disease classification and reliable remedy retrieval. The proposed component minimizes human error, reduces unnecessary chemical usage, and promotes sustainable plant-health management for Sri Lankan farmers.

Keywords: Leaf disease detection, Convolutional Neural Network, MobileNetV2, transfer learning, HSV segmentation, offline mobile inference, TensorFlow Lite, Flutter, remedy recommendation, smart agriculture.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DL	Deep Learning
FPS	Frames Per Second
GPU	Graphics Processing Unit
HSV	Hue, Saturation, Value Color Space
IPM	Integrated Pest Management
ML	Machine Learning
OpenCV	Open Source Computer Vision Library
ReLU	Rectified Linear Unit
SUS	System Usability Scale
TFLite	TensorFlow Lite
UI / UX	User Interface / User Experience
VGG16	Visual Geometry Group 16-layer Convolutional Neural Network

I. INTRODUCTION

1.1 Background Study and Literature Review

1.1.1 Background Study

Agriculture is one of the oldest and most important sectors for human survival and economic growth. In developing countries like Sri Lanka, agriculture is not only a major contributor to the national economy but also the primary livelihood for the majority of rural communities. Among the wide variety of crops cultivated across the country, plantation crops, fruit trees, and vegetable cultivars play a particularly important role, as they provide both nutrition for local consumption and income through domestic markets and exports. However, despite their economic importance, these crops face many challenges, with foliar disease infestation being one of the most critical and continuously recurring threats.

Foliar diseases such as Anthracnose, Red Rust (algal leaf spot), and Leaf Miner damage cause significant harm to a wide range of crops. Anthracnose, a fungal disease caused by species of Colletotrichum, produces dark sunken lesions on leaves and fruits and is particularly damaging to mango, banana, chilli, and several legumes. Red Rust, caused by the parasitic alga Cephaleuros virescens, appears as raised orange-red spots on tea, mango, citrus, and coffee, and reduces photosynthetic capacity over time. Leaf Miner damage, caused by the larvae of insects such as Liriomyza species, results in distinctive serpentine tunnels on the leaf surface that severely reduce plant vigour. Together, these three conditions are responsible for substantial yield loss across Sri Lanka's smallholder farming sector.

Figure 1: Foliar Disease Damage on Plant Leaves

Traditionally, farmers depend on manual inspection to identify diseases and assess the level of crop damage. While this method has been practised for decades, it suffers from several well-known limitations. Manual observation is time-consuming, less accurate, and heavily dependent on the experience of the individual farmer. In many cases, early signs of infection are missed or misidentified, which results in delayed responses and substantially larger crop damage by the time intervention is taken. Misidentification is particularly common between visually similar conditions — for example, between fungal anthracnose and bacterial leaf spots, or between algal red rust and ordinary nutrient-deficiency spots.

To overcome disease-related losses, farmers often apply chemical pesticides and fungicides without precise knowledge of the actual disease type or its severity. This excessive and untargeted usage not only increases production cost but also creates long-term problems such as soil degradation, water pollution, residue accumulation in food, and serious health risks for the farmers themselves. At the same time, global agricultural trends increasingly emphasize that sustainable farming requires reducing chemical inputs and adopting smarter, technology-driven decision-making.

Recent advances in Artificial Intelligence (AI) and Image Processing have created new and exciting opportunities in the agricultural sector. With the help of deep learning models — in particular Convolutional Neural Networks (CNNs) — it is now possible to classify diseases accurately from digital images of leaves. Similarly, color segmentation and contour detection

techniques allow automated measurement of infected leaf areas to determine the severity of an infection. Such methods provide farmers with a faster, more accurate, and more reliable decision-making system compared to traditional inspection.

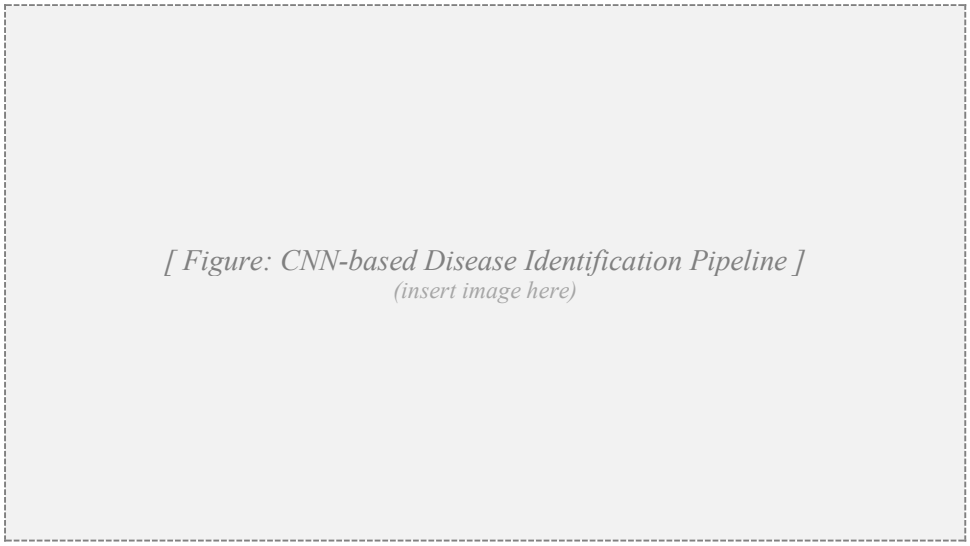

Figure 2: CNN-based Disease Identification Pipeline

The proposed Leaf Disease Identification and Remedy Recommendation System addresses these challenges. It combines a CNN-based transfer learning model — built on the lightweight MobileNetV2 backbone — for disease classification, together with HSV-based segmentation for image preparation and a structured remedy knowledge base for treatment guidance. The system identifies whether a given leaf is affected by Anthracnose, Red Rust, or Leaf Miner damage, and then returns culturally appropriate, organic, and chemical remedies for the identified condition. The backend is implemented in Python and the mobile application is built using Flutter, making the system accessible, practical, and farmer-friendly for use in the Sri Lankan agricultural context.

A defining characteristic of this component is its offline-first design. The trained CNN model is converted to TensorFlow Lite and packaged inside the application along with a local SQLite remedy database. This means the entire diagnostic pipeline — from image capture to disease prediction to remedy display — runs entirely on the user's device, without any dependency on internet connectivity. This is a critical capability for Sri Lankan rural farming areas, where mobile data is often unreliable, expensive, or completely unavailable.

By integrating modern IT solutions into agriculture in this way, this research highlights the possibility of achieving both higher productivity and environmental sustainability. It also demonstrates how intelligent systems can serve as digital assistants for farmers, improving decision-making and promoting smart agricultural practices in Sri Lanka.

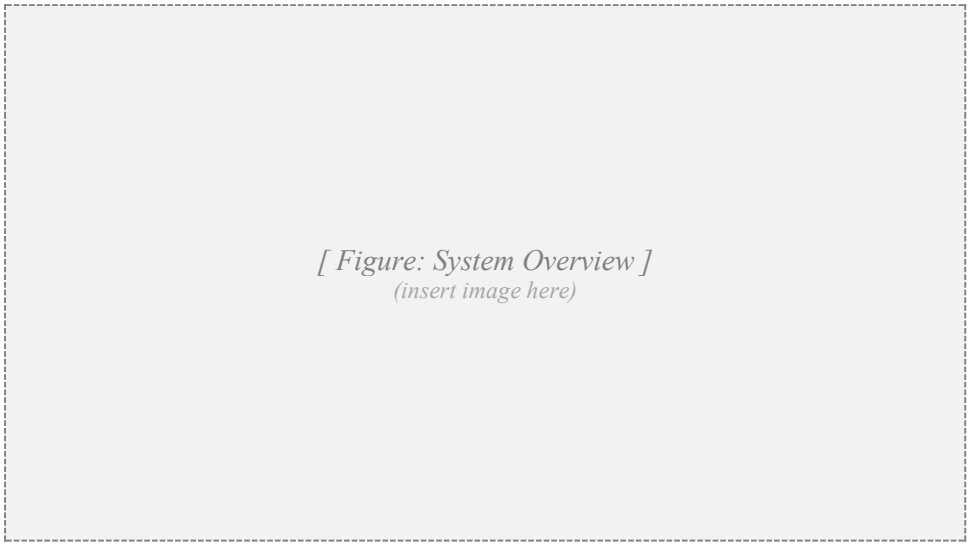

Figure 3: System Overview

1.2 Literature Review

Agriculture has always been a major research focus in the field of Information Technology, especially since the rise of Artificial Intelligence (AI) and Image Processing. Several researchers have attempted to solve the challenges of disease detection, crop monitoring, and treatment recommendation using a variety of technologies. This section reviews related works that guided the development of the proposed Leaf Disease Identification and Remedy Recommendation System.

In recent years, deep learning models — and Convolutional Neural Networks (CNNs) in particular — have been widely applied to plant disease identification. Mohanty et al. [1] used the PlantVillage dataset to demonstrate that CNN architectures such as AlexNet and GoogLeNet could classify 26 diseases across 14 crop species with over 99 percent accuracy on held-out test data. Their work established CNNs as a credible alternative to manual identification and inspired a large body of follow-up research. Similarly, Ferentinos [2] trained five different CNN architectures on

a dataset of 87,848 images and reported a top accuracy of 99.53%, again confirming that deep learning is well-suited to leaf-image classification.

Subsequent work has focused on three directions: improving generalization to field-captured (rather than laboratory) images, reducing model size for mobile deployment, and broadening the catalogue of supported diseases. Howard et al. [3] introduced the MobileNet family of architectures, which use depth-wise separable convolutions to dramatically reduce parameter count and inference latency. MobileNetV2, in particular, has become the de-facto choice for on-device plant-health applications because it offers a strong balance between accuracy and footprint. The proposed component follows this trajectory by selecting MobileNetV2 as its CNN backbone.

Apart from disease identification, several researchers have focused on leaf severity detection and area-of-infection measurement. Patil and Kumar [4] showed that color-based segmentation techniques such as HSV and LAB colour models can accurately measure infected regions of crop leaves, which is crucial for deciding the right type and dose of treatment. Singh et al. [5] combined image segmentation with classical machine learning classifiers to predict the progression of plant diseases, and their results suggest that the combination of deep-learning classification with image segmentation produces stronger overall decision-support than either approach alone.

Treatment recommendation is another important branch of the literature. Traditional advisory systems depend on soil testing and expert consultation, but newer IT-based systems link severity levels to treatment decision support. Wang et al. [6] proposed an automated advisory system that recommends suitable interventions based on the predicted disease and severity, aligning with the principles of precision agriculture, where inputs are applied selectively in order to reduce waste and environmental impact.

In the local Sri Lankan context, several studies have explored mobile-based applications for agriculture. Perera et al. [7] introduced a mobile advisory tool for paddy farmers that improved awareness about pest management practices. However, most of these systems are rule-based and do not incorporate AI-driven image processing. Furthermore, almost all existing tools require an active internet connection to perform meaningful analysis. This represents a significant shortcoming in rural settings where data connectivity is unreliable, and it identifies a clear gap that the proposed system aims to fill.

From the reviewed literature, it is clear that deep learning and image processing are becoming essential tools in modern agriculture. While many international studies have successfully demonstrated disease detection and severity analysis, only limited attempts have been made in Sri Lanka to integrate these technologies into farmer-friendly, offline-capable systems. The proposed Leaf Disease Identification and Remedy Recommendation System builds on these existing works but introduces a complete pipeline that covers disease classification, remedy recommendation, and offline mobile execution within a single platform. This makes the system more practical and applicable to real-world farming conditions in Sri Lanka.

1.3 Research Gap

Agriculture in Sri Lanka continues to face significant challenges due to foliar disease infestations across a wide range of cultivated crops. While many international studies have demonstrated success in applying Artificial Intelligence (AI) and Image Processing for plant disease detection, most published solutions are designed for large-scale farming systems in countries with advanced technological infrastructure and continuous internet connectivity. In Sri Lanka, however, the majority of growers — especially smallholder farmers in rural districts — still rely on manual visual inspection and experience-based decisions, which are not only time-consuming but also prone to misjudgement.

Existing mobile-based agricultural tools available locally are predominantly rule-based advisory systems that provide general recommendations but lack the ability to perform real-time, image-driven disease identification. Furthermore, even those few solutions that do incorporate machine learning typically depend on cloud inference, requiring a stable internet connection to upload the leaf image and receive a prediction. This reliance on connectivity is a fundamental barrier to adoption in rural Sri Lankan farming areas, where mobile data is often unreliable, costly, or simply unavailable for long stretches of the working day.

Another important gap lies in remedy guidance. Many existing tools stop at producing a disease label and leave the farmer to decide how to respond. Only a few tie the prediction directly to a curated, structured set of remedies covering cultural, organic, and chemical options. As a result, farmers often default to broad-spectrum chemical treatments, leading to long-term environmental damage, residue accumulation, and reduced soil fertility.

Therefore, there is a clear research gap in designing a unified solution that integrates:

1. Automated leaf disease identification using a CNN-based deep learning model.
2. Disease-specific remedy recommendation drawn from a curated knowledge base.
3. Fully offline operation, so that the entire pipeline runs on the user's device with no internet dependency.
4. A farmer-friendly mobile interface usable by individuals with limited technical experience.

This study addresses this gap by developing a Leaf Disease Identification and Remedy Recommendation System that combines a MobileNetV2-based CNN classifier, HSV image preprocessing, and an offline TensorFlow Lite deployment with a local SQLite remedy database, tailored specifically for the Sri Lankan farming context.

Feature / Capability	**Research [1]**	**Research [2]**	**Research [4]**	**Research [7]**	**Proposed System**
Disease identification using CNN / DL	✓	✓	✗	✗	✓
HSV-based image preprocessing	✗	✗	✓	✗	✓
Disease-specific remedy recommendation	✗	✗	✗	Partial	✓
Mobile application for real-time field use	✗	✗	✗	✓	✓
Fully offline inference (no internet required)	✗	✗	✗	✗	✓
Optional cloud sync for historical records	✗	✗	✗	✗	✓

Table 1: Research Gap Comparison

1.4 Research Problems

Crop cultivation, which forms a significant portion of the Sri Lankan rural economy, is continuously threatened by foliar disease infestations that reduce yield and increase production costs. Farmers currently depend largely on manual inspection and personal experience to identify

diseases and assess the resulting crop damage. This traditional approach suffers from several specific drawbacks:

- Inaccuracy: Farmers may fail to correctly distinguish between visually similar diseases — for example, anthracnose lesions can resemble bacterial spots, and red rust can be mistaken for nutrient deficiency. Limited specialist knowledge directly translates into incorrect treatment decisions.
- Delay in detection: Manual inspection of every leaf in a plot is impractical, and infections are typically noticed only after they have spread visibly. By that stage, severe damage may already have occurred and chemical intervention becomes both more expensive and less effective.
- Overuse of agrochemicals: When the disease is unclear, farmers commonly apply broad-spectrum fungicides or pesticides as a precaution. This practice damages soil fertility, contaminates groundwater, and harms beneficial insects.
- Reliance on connectivity: The few AI-based plant disease tools available on the market today require a continuous internet connection to function, which makes them unsuitable for remote and rural farming environments where mobile data coverage is poor or absent.
- Lack of integrated remedy guidance: Most existing tools either stop at the disease label or provide only generic advice, without offering structured cultural, organic, and chemical remedy options that the farmer can choose from.

Thus, the research problem can be defined as:

> *"How can a CNN-powered leaf disease identification and remedy recommendation system be designed to accurately detect Anthracnose, Red Rust, and Leaf Miner infections, and to provide farmers with sustainable, actionable treatment guidance entirely offline on a mobile device?"*

This research problem highlights the need for a solution that is accurate, farmer-friendly, sustainable, offline-capable, and able to reduce crop loss while promoting eco-friendly agricultural practices.

1.5 Research Objectives

Main Objective

To develop a CNN-powered Leaf Disease Identification and Remedy Recommendation System that integrates disease detection and disease-specific remedy guidance into a fully offline-capable mobile application, in order to assist Sri Lankan farmers in managing foliar disease infestations more effectively.

Specific Objectives

5. To design and implement a CNN-based transfer learning model — built on the MobileNetV2 backbone — for classifying paddy and crop leaf images into three target disease categories: Anthracnose, Red Rust, and Leaf Miner.
6. To collect and curate a balanced dataset of approximately 1,200 labelled images per disease class, totalling around 3,600 training samples drawn from public datasets, research databases, and field-collected images.
7. To apply HSV color segmentation and noise-reduction techniques during image preprocessing in order to improve classification accuracy under varied real-world lighting and background conditions.
8. To design and populate a remedy knowledge base that maps each predicted disease to a structured set of cultural, organic, and chemical interventions appropriate to that condition.
9. To package the trained model in TensorFlow Lite format and bundle it together with a local SQLite remedy database inside a Flutter mobile application, enabling fully offline diagnosis and remedy retrieval.
10. To integrate optional Firebase synchronisation so that historical diagnoses can be uploaded for officer review whenever an internet connection becomes available, without compromising offline core functionality.
11. To test and validate the system's accuracy in identifying the three target diseases, retrieving the correct remedy block, and operating reliably under real-world field conditions.

Business Objectives

12. To minimize crop losses caused by undetected foliar disease, thereby improving farmer productivity and household profitability.

13. To reduce the overuse of broad-spectrum chemical pesticides and fungicides, promoting environmentally friendly and sustainable farming practices in line with national agricultural policy.
14. To provide a low-cost, mobile-friendly, offline-capable solution that can be widely adopted by Sri Lankan farmers without requiring advanced technical knowledge or continuous internet access.
15. To contribute towards the digital transformation of agriculture in Sri Lanka, aligning with global trends in precision farming and smart agriculture.

2. METHODOLOGY

2.1 Methodology

The development of the Leaf Disease Identification and Remedy Recommendation System followed a structured methodology to ensure accurate disease classification, reliable remedy retrieval, and dependable offline operation. The project began with data collection, where leaf images for three disease classes — Anthracnose, Red Rust, and Leaf Miner — were gathered from multiple sources including farmer submissions, public datasets such as PlantVillage, and online plant pathology repositories. Approximately 1,200 images were collected for each class, giving a total of around 3,600 labelled samples. Each image was manually verified to confirm the correct disease label, and a parallel knowledge base was prepared to map each disease class to a structured set of cultural, organic, and chemical remedy options.

Once the dataset was prepared, image preprocessing was performed using OpenCV. All images were resized to 224×224 pixels to match the input requirements of the CNN model. Noise-reduction techniques, including Gaussian blur and median filtering, were applied to enhance image quality. HSV color segmentation was used to differentiate diseased regions from the background and from healthy tissue. To improve model generalization across varied field conditions, data augmentation was applied through rotation, horizontal and vertical flipping, scaling, and brightness adjustments.

For model development, MobileNetV2 was selected as the CNN backbone because it offers an excellent balance between classification accuracy and on-device inference efficiency. The pretrained ImageNet weights were loaded with the top classification layers removed, allowing fine-tuning for the specific three-class disease task. Additional layers included a GlobalAveragePooling2D layer, a dense layer with 128 neurons and ReLU activation, a dropout layer with a rate of 0.5 to prevent overfitting, and a final dense softmax layer with three output units. The model was trained for approximately 30 epochs with a batch size of 32, using the Adam optimizer with an initial learning rate of 1×10^{-3} and categorical cross-entropy as the loss function. EarlyStopping and ModelCheckpoint callbacks were employed to prevent overfitting and to save the best-performing weights. Performance metrics including accuracy, precision, recall, and F1-score were monitored on a held-out test set.

System integration involved connecting the trained model with a Python backend using TensorFlow/Keras for training and OpenCV for preprocessing. Once training was complete, the final model was exported in TensorFlow Lite format so that it could run directly on the user's mobile device. A mobile application was developed using Flutter to provide a user-friendly interface for farmers; users can capture or upload an image of an affected leaf, receive a disease classification result, and view the corresponding remedy block. A local SQLite database, bundled inside the application, stores the remedy knowledge base so that the entire diagnostic pipeline operates without any internet connection. Optional Firebase integration is provided so that diagnostic history can be synchronized to the cloud whenever connectivity becomes available.

Finally, the system underwent thorough evaluation and testing. Disease classification accuracy was verified using confusion matrices and comparison with expert-labelled images. Inference latency was measured on representative mid-range Android devices to confirm that the offline pipeline produces results within an acceptable time frame. User feedback was collected from farmers and agricultural officers, and the System Usability Scale (SUS) was used to assess the practicality of the application. The methodology also included technical, economic, operational, and scheduling feasibility assessments to ensure that the system could be implemented efficiently and effectively for real-world conditions. Overall, this methodology provided a comprehensive end-to-end approach, integrating data collection, preprocessing, model development, system integration, and evaluation to deliver a robust offline-capable Leaf Disease Identification and Remedy Recommendation System.

2.1.1 Agile Principles Applied in the Project

The development of the Leaf Disease Identification and Remedy Recommendation System followed Agile principles to ensure flexibility, continuous improvement, and effective team collaboration. One key principle applied was Adaptive Planning. The project was divided into multiple sprints, each focusing on specific tasks such as data collection, image preprocessing, model training, mobile integration, and offline deployment. Plans were updated at the end of each sprint based on testing results and stakeholder feedback. This allowed the team to adapt quickly to unexpected challenges, such as dataset imbalance, model accuracy plateaus, or unexpected behaviour during on-device inference, without affecting overall project progress.

Continuous Improvement was another important principle applied throughout the project. After each sprint, the team reviewed results — including model accuracy on the validation set, inference latency on real mobile devices, and feedback from farmers and agricultural officers. Adjustments were then made to preprocessing techniques, hyperparameters, model architecture choices, and the mobile interface design. This iterative approach ensured that each new version of the system was better than the previous one, gradually improving accuracy, efficiency, and user satisfaction.

Risk Management was actively integrated into the Agile process. Potential risks such as mislabelling of training images, the trained model being too large to fit comfortably inside a mobile application, delays in mobile deployment, or insufficient image diversity were identified early and monitored throughout the project. Mitigation strategies included cross-validation of the dataset, conversion of the trained model to TensorFlow Lite for size reduction and faster inference, supplementing the dataset through aggressive augmentation, and regular progress tracking during sprint reviews. By continuously monitoring risks, the team minimized the chances of critical issues affecting project delivery.

Collaboration played a vital role in ensuring project success. Team members, including developers, researchers, and agricultural advisers, worked closely to exchange knowledge and provide feedback. Regular sprint meetings allowed the team to discuss challenges, share progress, and coordinate tasks effectively. Collaboration also extended to stakeholders such as farmers and instructors, whose input directly guided improvements in usability, practicality, and real-world applicability of the system.

Finally, the Agile approach contributed to Improved Team Morale. By breaking the project into manageable sprints, celebrating small successes, and incorporating stakeholder feedback at every step, team members were motivated and engaged throughout the development process. The iterative workflow and the collaborative environment helped maintain focus, reduce stress, and encourage ownership of tasks, ultimately leading to a high-quality and practical Leaf Disease Identification and Remedy Recommendation System.

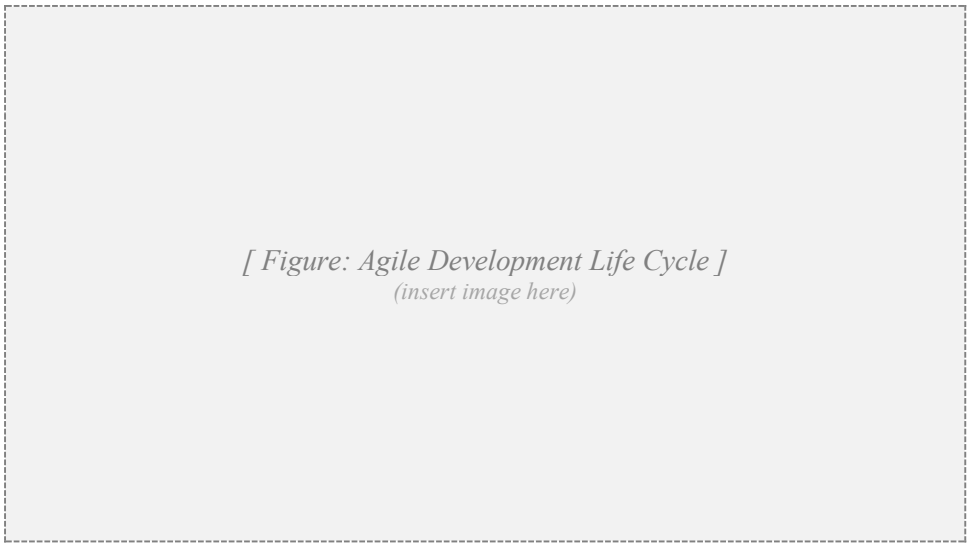

Figure 4: Agile Development Life Cycle

2.1.2 Feasibility Study and Planning

Before starting the development of the Leaf Disease Identification and Remedy Recommendation System, a feasibility study was conducted to ensure the project could be successfully implemented. Technical feasibility was the first consideration. The system required robust image processing and deep learning capabilities for disease identification, together with on-device inference to support offline use. Tools such as OpenCV for image preprocessing, MobileNetV2 (via TensorFlow/Keras) for transfer learning, TensorFlow Lite for mobile deployment, Flutter for cross-platform mobile development, and SQLite for local remedy storage were assessed. The combined availability of these mature open-source technologies made the system technically feasible. Hardware requirements were also moderate, since the conversion of MobileNetV2 to TensorFlow Lite produces a small model file (approximately 9 MB) that runs efficiently on standard mid-range Android and iOS smartphones.

Economic feasibility was another critical factor. The project relied primarily on open-source software libraries and freely available pretrained model weights, which significantly reduced development cost. The use of existing image datasets, supplemented by farmer-contributed and field-collected images, minimized the need for expensive data acquisition. Optional Firebase synchronisation uses a free tier sufficient for small-scale pilot deployment. By avoiding costly proprietary solutions, the system could be delivered as a low-cost tool accessible to small- and medium-scale farmers in Sri Lanka, ensuring affordability without compromising performance.

Operational feasibility focused on the practical application of the system. The design of the mobile app prioritized a user-friendly interface to allow farmers with minimal technical experience to upload images, view disease classifications, and obtain remedy recommendations in just a few taps. Critically, because all inference occurs on-device, farmers in remote rural areas can benefit from the system even when no mobile signal is available. Training sessions and brief illustrated guidance materials were also planned to support effective system use during initial rollout.

Finally, scheduling feasibility was considered to manage project development efficiently. The work was divided into clearly defined phases including data collection, preprocessing, model training, system integration, mobile deployment, offline testing, and final validation. Each phase was scheduled as a separate sprint within the Agile framework, allowing iterative progress and flexibility to accommodate delays or unexpected issues. By breaking the project into manageable tasks and reviewing progress regularly, the team ensured that the system could be completed within the allocated timeframe while maintaining high-quality outcomes.

In summary, the feasibility study confirmed that the Leaf Disease Identification and Remedy Recommendation System was technically possible, economically viable, operationally practical, and schedulable within the project timeframe. This planning phase provided a clear roadmap for development and ensured that the system could be effectively implemented to support sustainable farming in Sri Lanka.

Gantt Chart

[Figure: Gantt Chart of Sprint Schedule]
(insert image here)

Figure 5: Gantt Chart

The Gantt chart above illustrates the sprint-by-sprint development schedule of the component, beginning with planning and research and progressing through data collection, preprocessing, model development and training, integration with the remedy recommendation module, mobile application integration, final testing and refinement, and the final launch and monitoring phase.

Work Breakdown Chart

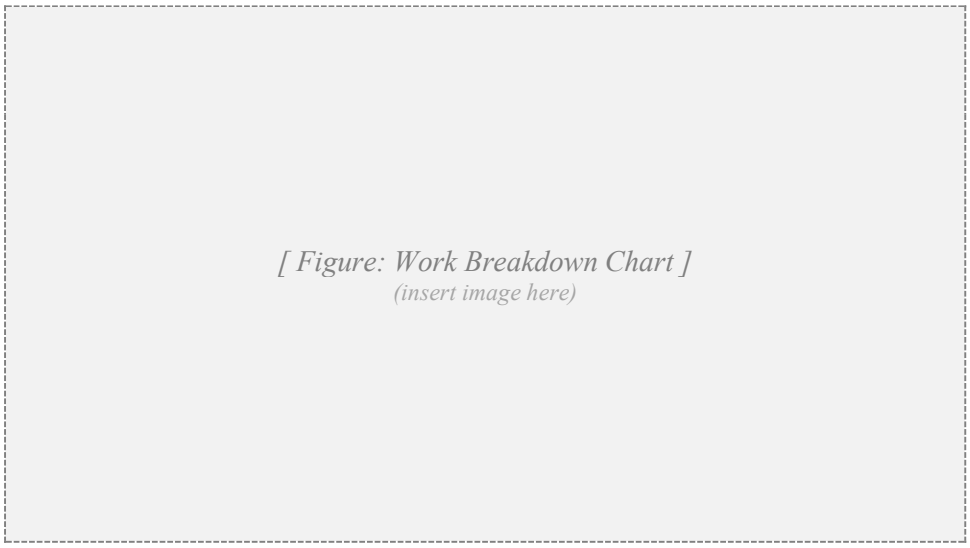

Figure 6: Work Breakdown Chart

The Work Breakdown Chart above decomposes the component-level project into its constituent activities, grouped under Initialization, Planning, Design, Implementation, Testing, and Launch phases. This breakdown helped distribute responsibilities and clarify dependencies during execution.

2.1.3 Requirement Gathering & Analysis

Functional Requirements

- The system must automatically classify leaf images into one of three disease categories — Anthracnose, Red Rust, or Leaf Miner — and provide farmers with clear guidance on which condition has been detected.
- The system should accept input from either the device camera or the device gallery, and apply HSV-based image preprocessing to clean the image before passing it to the CNN model.

- Once a disease is identified, the system must retrieve the corresponding remedy block from a local SQLite knowledge base and display cultural, organic, and chemical treatment options.
- The system must operate fully offline: image capture, preprocessing, inference, and remedy display must all complete without any internet connection.
- The system must allow users to view a local history of past diagnoses, including the captured image, predicted disease, confidence score, and timestamp.
- The system should generate a printable or shareable diagnostic report that summarizes the disease, confidence, and recommended remedies for easy reference.
- When a network connection is available, the system should optionally synchronise diagnostic history to Firebase so that agricultural officers can review aggregated data.

Non-Functional Requirements

- The system must maintain an overall classification accuracy of at least 90% on the held-out test set, in order to be reliable enough for field use.
- End-to-end response time on a mid-range Android device — from image capture to remedy display — must remain under 2 seconds.
- The mobile application must be designed to be user-friendly and intuitive, requiring minimal training for effective use even by farmers with limited technical knowledge.
- All user data — including stored images and diagnostic history — must be retained securely on the device, with optional encrypted synchronisation to Firebase when online.
- The system must be scalable so that additional disease classes, alternative remedies, or new languages can be added without redesigning the core architecture.
- The bundled model and remedy database together must remain under 25 MB to minimize the application install size.

User Requirements

- Farmers require a simple and accessible mobile interface that allows them to capture or upload a leaf image, receive an accurate disease classification, and obtain practical remedy guidance — all in a few taps.

- Agricultural officers and researchers need access to historical diagnostic data and analytical reports to support advisory work and to monitor regional disease trends.
- All users expect the system to provide accurate, timely, and actionable information that supports real-world decision-making in plant-health management.
- The mobile application must support offline usage so that farmers can capture images and receive complete diagnoses in the field, with results being optionally synchronized when connectivity is later restored.

System Requirements

- The training and conversion backend is implemented in Python, with TensorFlow/Keras for model development and OpenCV for image preprocessing.
- Mobile deployment uses Flutter, enabling cross-platform compatibility for both Android and iOS while preserving a consistent user experience.
- The trained CNN model is converted to TensorFlow Lite format and bundled inside the application, ensuring fast and efficient on-device inference without requiring a network connection.
- A local SQLite database stores the remedy knowledge base, allowing the application to look up disease-specific remedies entirely offline.
- Optional Firebase integration provides cloud-based storage of diagnostic history when connectivity is available, supporting future analytics and officer-level review.
- The system must be capable of supporting multiple users concurrently when synchronising to Firebase, without significant performance degradation.

Hardware Requirements

- The mobile application should run on Android or iOS smartphones with at least 2 GB of RAM and a basic processor capable of handling TensorFlow Lite inference efficiently.
- A development workstation equipped with a GPU is recommended for training the deep learning model, in order to accelerate the training of the CNN over the full dataset.
- Internet access is required only during optional cloud synchronisation; the core diagnostic pipeline does not require connectivity.

- Optional accessories such as a phone macro lens or improved smartphone camera can enhance image quality, but are not strictly required for basic operation.

Use Case Scenarios

Use Case ID	UC-01
Use Case Name	Leaf Disease Diagnosis and Remedy Recommendation
Preconditions	1. The farmer has the mobile app installed and opened. 2. The mobile device has a working camera or an existing image in its gallery. 3. Internet connectivity is NOT required for diagnosis.
Primary Actor	Farmer / Field User
Main Success Scenario	1. The farmer opens the mobile application. 2. The farmer captures or selects an image of an affected leaf. 3. The system preprocesses the image using OpenCV (resize, denoise, HSV segmentation). 4. The MobileNetV2-based CNN classifies the image into one of the three disease classes. 5. The system retrieves the corresponding remedy block from the local SQLite database. 6. The farmer is shown the disease name, confidence score, and a structured list of cultural, organic, and chemical remedies. 7. The result is saved to the local diagnostic history.
Extensions	6a. The farmer can revisit historical diagnoses at any time from the local history view. 6b. If an internet connection becomes available, diagnoses can be synchronised to Firebase for officer-level review. 6c. If the captured image is unclear, the system warns the user and requests a re-capture.

Table 2: Use Case from the Farmer

User Stories

User Story 1 – Farmer:

As a farmer, I want to upload or capture images of leaves from my field directly on my phone so that I can quickly identify whether the plant is affected by Anthracnose, Red Rust, or Leaf Miner, and receive practical remedy options that I can apply on the same day.

Acceptance Criteria:

- The farmer can take or upload a photo of a leaf using a few taps.
- The system classifies the disease accurately into one of the three target classes.
- The disease label is displayed clearly along with a confidence score.

- Remedies are shown grouped into cultural, organic, and chemical options.
- The entire interaction works without an internet connection.

User Story 2 – Agricultural Officer:

As an agricultural officer, I want to access historical disease diagnoses recorded by farmers in my region so that I can monitor outbreaks, advise farmers more effectively, and support sustainable plant-health management.

Acceptance Criteria:

- Officer can view aggregated reports of disease occurrences once farmers' devices have synced to Firebase.
- Officer can filter data by date, location, or disease type.
- Reports include actionable insights for plant-health advisory work.
- Data is accessible via the same mobile or a companion web interface.

2.1.4 Designing the System

The design of the Leaf Disease Identification and Remedy Recommendation System focuses on providing farmers with a practical, easy-to-use, and reliable tool for detecting foliar diseases and obtaining clear treatment guidance. The system was designed to be modular, with separate components for image acquisition, preprocessing, disease classification, remedy retrieval, and mobile reporting. This modular approach ensures that each part of the system can function independently and be upgraded or expanded in the future without disrupting the rest of the pipeline.

During the design phase, a user-centred approach was adopted. The system is intended to support farmers who may have limited technical knowledge, so the workflow was kept deliberately simple: the user captures or uploads an image of an affected leaf, and the system automatically identifies the disease and presents the corresponding remedy block. The mobile application, developed using Flutter, serves as the primary interface, while the supporting backend (used for training and future model updates) is implemented in Python with TensorFlow/Keras and OpenCV. A local SQLite database stores the remedy knowledge base, ensuring that the diagnostic pipeline operates entirely on-device. Firebase is used as an optional cloud sync layer for diagnostic history.

The system design emphasizes accuracy, speed, and usability. The image preprocessing module resizes and denoises images to enhance model performance. The disease classification module, based on MobileNetV2 transfer learning, predicts the disease class and produces a confidence score. The remedy retrieval module looks up the predicted class in the local SQLite database and returns a structured set of cultural, organic, and chemical remedies. The mobile reporting module displays the result clearly to the farmer and persists it to the local diagnostic history.

Overall, the design phase aimed to create a robust, efficient, and farmer-friendly system that combines AI, image processing, and offline mobile technology to support sustainable plant-health management, reduce crop loss, and minimize unnecessary chemical input.

2.1.5 System Architecture Diagram

The System Architecture of the Leaf Disease Identification and Remedy Recommendation System illustrates how different components interact to provide a seamless experience for the user. The architecture is layered, combining the mobile application interface, image processing modules, the

on-device deep learning model, the local remedy database, and an optional cloud synchronisation layer.

At the user layer, farmers interact with the system through a Flutter-based mobile application. This layer allows users to capture or upload images of affected leaves. Once the image is selected, it is sent to the preprocessing layer, implemented using OpenCV, where images are resized to 224×224 pixels, denoised, and segmented using HSV color space to highlight diseased regions. Data normalisation ensures that the resulting image is compatible with the deep learning model and helps stabilise prediction quality.

The model inference layer contains the MobileNetV2-based CNN, which was trained on the three target disease classes and then exported to TensorFlow Lite format. This layer predicts the most likely disease and produces a probability distribution over the three classes. After classification, the remedy retrieval layer queries the local SQLite database using the predicted class as a key. The corresponding remedy block — covering cultural, organic, and chemical interventions — is returned to the application and displayed to the user.

All diagnostic results, including the captured image thumbnail, the predicted disease, the confidence score, and the timestamp, are stored locally in the device's diagnostic history. Whenever an internet connection is available, this history can be optionally synchronised to the Firebase cloud storage layer, enabling agricultural officers and researchers to access aggregated regional data and generate trend reports. The modular architecture ensures scalability, allowing new disease classes, additional remedy options, or alternative model architectures to be incorporated without disrupting the existing workflow.

Figure 7 presents the system architecture diagram, showing the data flow from the mobile app to preprocessing, model inference, remedy retrieval, local storage, and optional cloud synchronisation. Arrows indicate the sequence of data processing, while each block represents a functional module. This diagram provides a clear overview of how the system integrates AI, image processing, on-device storage, and optional cloud technologies to support sustainable plant-health management.

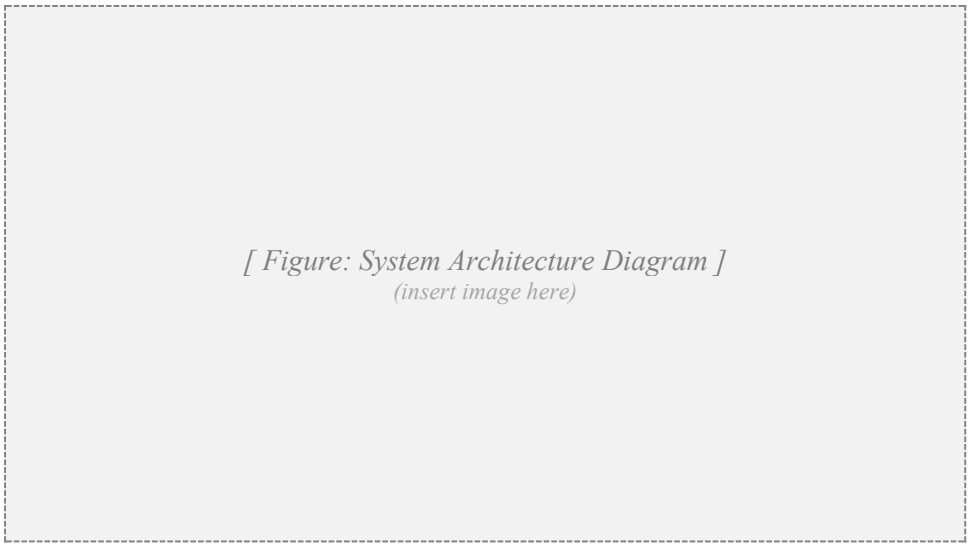

Figure 7: System Architecture Diagram

2.1.6 Implementation

The implementation phase focuses on transforming the system design into a working solution that accurately identifies the three target diseases, retrieves disease-specific remedies, and delivers everything through a smooth offline-capable mobile interface. The system was implemented using Python for the model-training backend, TensorFlow/Keras for the deep learning model, OpenCV for image preprocessing, Flutter for the mobile application, SQLite for local remedy storage, and Firebase for optional cloud synchronisation.

The workflow begins inside the mobile app, where the farmer either captures a fresh image of a leaf or selects one from the device's gallery. The image is then passed to the on-device preprocessing module, which resizes it to 224×224 pixels, applies a Gaussian blur to remove noise, and performs HSV-based segmentation to suppress background distractions. Data augmentation techniques such as rotation, flipping, and brightness adjustment were applied during training to improve the robustness of predictions, but inference itself uses only the cleaned, normalized version of the input image.

The disease classification module is built on the MobileNetV2 transfer learning model, pretrained on ImageNet. The top layers of MobileNetV2 were replaced with a GlobalAveragePooling2D layer, a Dense layer with 128 neurons and ReLU activation, a Dropout layer with rate 0.5 to reduce overfitting, and a final softmax output layer with three units corresponding to the three disease classes. During training, EarlyStopping and ModelCheckpoint callbacks were used to retain the

best weights and avoid overfitting. The trained model was then converted to TensorFlow Lite for deployment on mobile devices.

The remedy retrieval module is implemented as a local SQLite database that ships with the application. Each row in the remedies table corresponds to one disease class, and stores cultural, organic, and chemical remedies as separate fields. After the CNN has produced a prediction, the application uses the predicted class label as a primary key to look up the appropriate row and render the structured remedy block to the user.

All outputs, including disease name, confidence score, and the corresponding remedy block, are written to the local diagnostic history. When an internet connection is available, the application can synchronise this history to Firebase, allowing agricultural officers to view aggregated diagnostic data. During the implementation phase, unit testing was performed on each module, followed by integration testing to ensure that all components worked seamlessly together. The implementation phase successfully transformed the system from design to a functional prototype, combining AI, image processing, and offline mobile technology to support real-time plant-health decision-making.

Figure 8: Mobile Interface (Disease Diagnosis Screen)

[Figure: Image Preprocessing and HSV Segmentation Code]
(insert image here)

Figure 9: Image Preprocessing and HSV Segmentation

[Figure: Inference from Saved Model and Output Visualization]
(insert image here)

Figure 10: Model Inference and Output Visualization

[Figure: MobileNetV2 Model Architecture and Transfer Learning Layers]
(insert image here)

Figure 11: MobileNetV2 Model Architecture and Transfer Learning Layers

2.1.7 Development Environment and Tools

The development of the Leaf Disease Identification and Remedy Recommendation System utilized a combination of software and hardware tools selected to ensure accuracy, efficiency, and offline reliability. The model-training backend was implemented using Python, which provided a flexible platform for integrating deep learning, image processing, and supporting libraries. The deep learning model, based on MobileNetV2 transfer learning, was developed using TensorFlow and Keras, enabling fast experimentation and reliable training.

Image preprocessing and segmentation were performed using OpenCV, which allowed efficient manipulation of images, noise reduction, and HSV-based color segmentation. Data augmentation techniques such as rotation, flipping, zoom, and brightness adjustments were applied using TensorFlow's ImageDataGenerator, enhancing the model's ability to generalize under varied field conditions including different lighting, leaf orientations, and partial occlusion.

For the mobile application, Flutter was chosen to provide a cross-platform solution, allowing the app to run seamlessly on both Android and iOS devices. The trained CNN was converted to TensorFlow Lite to enable fast on-device inference. The remedy knowledge base is stored in a bundled SQLite database, ensuring that the entire diagnostic experience operates without an internet connection. Firebase is used optionally for cloud synchronisation of diagnostic history, but is never required for core functionality.

The hardware environment for model development and testing included a development PC equipped with a GPU-enabled graphics card to accelerate deep learning training. Mobile testing was performed on Android smartphones with a minimum of 2 GB RAM to confirm smooth performance of the Flutter application and acceptable real-time inference latency.

Summary of tools and libraries:

- Python 3.10+ – Backend programming language for training and conversion.
- TensorFlow / Keras – Deep learning framework used to build and train the MobileNetV2-based CNN.
- OpenCV – Image preprocessing and HSV-based segmentation.
- Flutter (Dart) – Cross-platform mobile application development for Android and iOS.
- TensorFlow Lite – Model deployment runtime for on-device inference.
- SQLite (sqflite plugin) – Local storage of the remedy knowledge base on the user's device.
- Firebase – Optional cloud synchronisation for diagnostic history.
- GPU-enabled workstation – Used during model training and experimentation.

The combination of these tools and technologies provided a robust, scalable, and user-friendly system that integrates AI-based disease identification, structured remedy recommendation, and fully offline mobile operation, making it suitable for real-world deployment in Sri Lankan farming.

2.1.8 Data Collection

The data collection phase is a critical part of developing the Leaf Disease Identification and Remedy Recommendation System, as the quality and diversity of the dataset directly influence the accuracy of the CNN model. The dataset comprises images of three target disease classes — Anthracnose, Red Rust, and Leaf Miner — commonly observed on a variety of crops grown in Sri Lanka, including mango, banana, citrus, tea, vegetables, and ornamental plants. Each class contains approximately 1,200 labelled images, totalling around 3,600 samples, covering different stages of infection, lighting conditions, and leaf orientations.

Images were collected from three main sources:

16. Public datasets: Open-access collections including PlantVillage and various Kaggle plant-disease repositories were used as the primary source of laboratory-quality images for each disease class.
17. Field photography: Original field images were captured in real cultivation areas using smartphones, in order to introduce the natural variability that the model will encounter during real use.
18. Web-scraped images: Additional images were collected from agricultural extension websites and verified manually for label correctness before being added to the dataset.

Along with the disease label, each image was also tagged with a quality flag indicating whether it represented a clear, partial, or borderline case of the disease. This auxiliary tagging supported subsequent analysis of misclassification patterns.

To support training, the dataset was organised into training and testing folders, maintaining a balanced class distribution. Metadata such as image source, capture date, and (where available) approximate location was also recorded to enable future studies and improvements in model performance.

The collected dataset forms the foundation for data preprocessing, model training, and evaluation, ensuring that the system can accurately classify the three target diseases and provide reliable, actionable remedy guidance under real-world conditions.

[Figure: Dataset Folder Structure (Disease Classes)]
(insert image here)

Figure 12: Disease Classes (Dataset Folders)

[Figure: Sample Disease Images from the Dataset]
(insert image here)

Figure 13: Sample Disease Images from the Dataset

2.1.9 Data Preprocessing

Data preprocessing is a crucial step in the Leaf Disease Identification and Remedy Recommendation System, as it ensures that the input images are clean, standardised, and suitable for training the deep learning model. The raw images collected from public datasets, field photography, and web sources often contain noise, varying sizes, inconsistent lighting conditions, and different orientations, all of which can negatively impact model performance if not properly handled.

[Figure: Data Preprocessing Code (HSV Segmentation)]
(insert image here)

Figure 14: Data Preprocessing Code

2.1.9.1 Image Resizing and Normalization

All images were resized to 224×224 pixels — the input size expected by MobileNetV2 — ensuring consistent dimensions across the dataset. Pixel values were normalised to the 0–1 range by dividing each channel by 255. This normalisation step stabilises gradient updates during training and helps the model converge faster and more reliably.

2.1.9.2 Data Augmentation

To enhance model robustness and prevent overfitting, data augmentation was applied using TensorFlow's ImageDataGenerator. The augmentation techniques included:

- Rotation (±25 degrees)
- Horizontal and vertical flipping
- Zooming in or out by up to 15%
- Brightness adjustments to simulate varied lighting conditions
- Width and height shifts of up to 10%

These transformations help the model learn features that are invariant to orientation, scale, and lighting variation, mimicking the kinds of variation the model will see in real-world field photographs.

2.1.9.3 Noise Reduction

Images captured in natural environments often include background noise, shadows, or compression artifacts. Gaussian blur and median filtering were applied using OpenCV to remove unwanted noise while preserving important leaf and lesion features.

2.1.9.4 Leaf Segmentation Using HSV Masking

To improve focus on the diseased region of each leaf, every image was converted from RGB to HSV color space. A healthy leaf mask was created by defining thresholds for hue, saturation, and value that capture typical green leaf tissue. Inverting this mask highlighted the affected (non-green) regions of the leaf, which were further processed using morphological operations such as opening and closing to remove small noise blobs and refine the contours of the diseased area. While the

trained model receives the full preprocessed image as input, this segmentation step is also used independently to visualise affected areas to the farmer in the diagnostic report.

2.1.9.5 Dataset Splitting

The preprocessed dataset was split into training, validation, and testing sets in a 70:15:15 ratio, ensuring balanced representation of all three disease classes in each subset. This splitting allows the system to be trained on diverse samples while being evaluated honestly on data it has never seen during training.

These preprocessing steps ensured that the dataset fed into the MobileNetV2 model is consistent, clean, and representative of real-world conditions. Proper preprocessing improves model accuracy, reduces misclassification, and supports reliable disease identification — which in turn drives correct remedy recommendation.

2.1.10 Model Training

The model training phase focuses on teaching the CNN to accurately classify leaf images into the three target disease classes — Anthracnose, Red Rust, and Leaf Miner. The system leverages transfer learning with the MobileNetV2 model, pretrained on ImageNet, which allows the model to benefit from features learned on a large general-purpose image dataset and adapt them to disease classification with a relatively modest amount of domain-specific data.

2.1.10.1 Model Architecture

The pretrained MobileNetV2 model was loaded without its top classification layers (include_top=False) so that custom layers could be added for the specific three-class task. The following layers were stacked on top of the frozen base:

- GlobalAveragePooling2D — Aggregates spatial features into a fixed-length feature vector suitable for classification.
- Dense Layer (128 neurons, ReLU activation) — Learns high-level representations specific to leaf-disease imagery.
- Dropout Layer (rate = 0.5) — Prevents overfitting by randomly disabling 50% of neurons during each training step.
- Softmax Output Layer (3 units) — Produces probabilities for the three disease classes.

All base MobileNetV2 layers were frozen during the initial training phase to preserve the pretrained features, while only the top layers were trained on the disease dataset. In a subsequent fine-tuning phase, the upper convolutional blocks of MobileNetV2 were unfrozen and trained at a smaller learning rate to further specialise the feature extractor on disease-specific patterns.

2.1.10.2 Training Parameters

The model was trained using the following parameters:

- Optimizer: Adam, chosen for its fast convergence and adaptive learning rate.
- Initial Learning Rate: 1×10^{-3}, reduced on plateau.
- Loss Function: Categorical cross-entropy, suitable for multi-class classification.
- Batch Size: 32
- Epochs: 25–30, controlled by early stopping.
- Callbacks:
- EarlyStopping — Monitored validation loss and stopped training if no improvement occurred over 5 consecutive epochs.
- ModelCheckpoint — Saved the best model weights based on validation accuracy.
- ReduceLROnPlateau — Lowered the learning rate when validation loss plateaued.

Training was performed on a GPU-enabled PC to accelerate computation. The dataset was fed using ImageDataGenerator, which performed on-the-fly data augmentation to improve generalisation.

2.1.10.3 Training Process and Evaluation

During training, both training and validation accuracy were continuously monitored. Early stopping prevented overfitting, and the final model achieved approximately 92% validation accuracy across the three disease classes. The macro-averaged F1-score across the three classes reached 0.92, indicating high reliability across all categories rather than accuracy concentrated in a single dominant class.

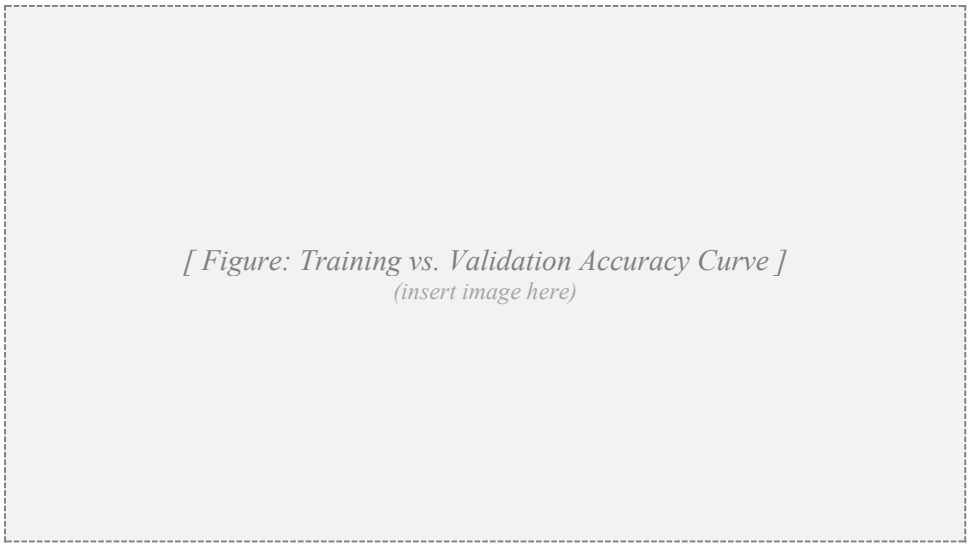

Figure 15: Trained Model Accuracy

After training, the model was saved as a .h5 file and then converted to TensorFlow Lite format for mobile deployment. The resulting .tflite file is approximately 9 MB in size, allowing it to be bundled directly inside the mobile application without bloating the install size. This enables real-time disease detection on the Flutter-based mobile application, providing farmers with quick and accurate results entirely offline.

The model training process successfully combined transfer learning, data augmentation, and modern optimisation techniques to create a robust disease classification system. By carefully monitoring validation performance and using early stopping, the system avoids overfitting while providing reliable predictions, forming the backbone of the remedy recommendation pipeline.

2.1.11 How the Algorithms Work

The Leaf Disease Identification and Remedy Recommendation System relies on a combination of deep learning and image processing algorithms to detect diseases, prepare images for inference, and provide structured remedy guidance. The algorithms operate as a pipeline, ensuring seamless processing from a raw camera image to actionable output for the farmer.

2.1.11.1 Disease Identification Algorithm

The disease identification algorithm uses MobileNetV2-based transfer learning for image classification. The process includes the following steps:

19. Input Preprocessing: The uploaded or captured image is resized to 224×224 pixels and normalised so that pixel values lie in the range 0–1.
20. Feature Extraction: The frozen convolutional layers of MobileNetV2 extract high-level features from the leaf, such as lesion shape, color distribution, edge texture, and tunnel patterns.
21. Classification: The custom dense layers process these features and output a probability distribution over the three disease classes using the softmax activation function.
22. Prediction: The class with the highest probability is selected as the predicted disease, and the associated confidence score is returned alongside it.

2.1.11.2 HSV-Based Image Preparation Algorithm

In parallel with classification, an HSV-based preparation pass is performed to support visual feedback to the user:

23. HSV Conversion: The input image is converted from RGB to HSV color space to separate hue (the color of the pixel) from saturation and brightness.
24. Healthy Leaf Masking: A healthy leaf mask is generated using thresholds on hue, saturation, and value that capture typical green tissue.
25. Affected Region Extraction: The mask is inverted to highlight the affected (non-green) regions of the leaf.
26. Morphological Cleanup: Opening and closing operations remove small noise blobs and refine the contours of the diseased area.
27. Visual Overlay: The detected affected regions are drawn back onto the original image to give the farmer a clear visual confirmation of where the system is observing damage.

2.1.11.3 Remedy Recommendation Algorithm

Based on the predicted disease, the system retrieves the corresponding remedy block:

- Anthracnose → Cultural sanitation, neem oil / Bordeaux mixture, Mancozeb / Carbendazim.

- Red Rust → Pruning and ventilation, Bordeaux mixture / lime-sulphur, Copper Oxychloride.
- Leaf Miner → Removal of mined leaves, neem oil / Beauveria bassiana, Imidacloprid / Abamectin.

This mapping is stored in a local SQLite database that ships with the application, ensuring quick and consistent recommendations without any network call.

The complete algorithmic workflow can be visualised as:

Mobile App → Image Preprocessing → MobileNetV2 Disease Classification → SQLite Remedy Lookup → Result Display → (Optional) Firebase Sync

By combining deep learning for disease identification with image processing for visual feedback and a structured local knowledge base for remedy retrieval, the system achieves accurate, fully offline, real-time results, enabling farmers to take informed actions for disease control and crop protection.

Figure 16: MobileNetV2 / CNN Layer Architecture

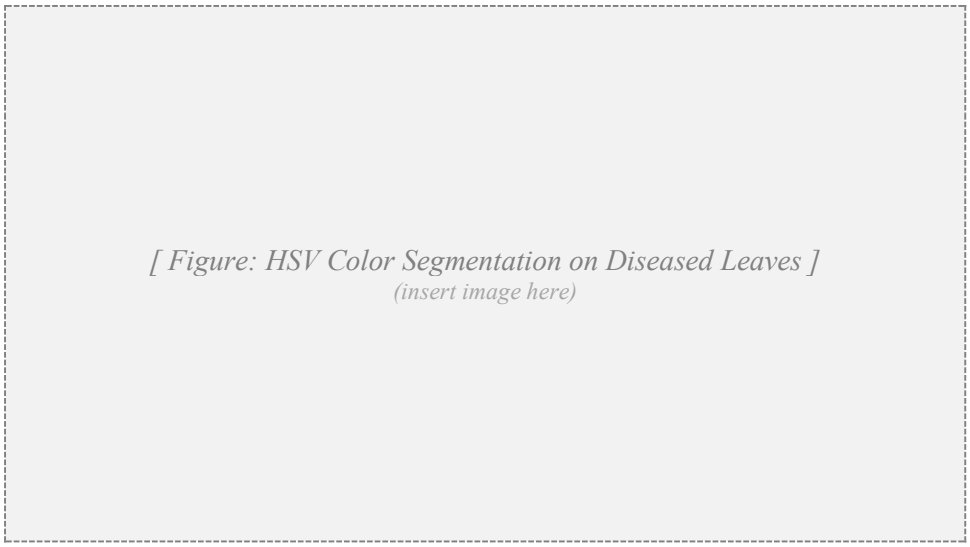

Figure 17: HSV Color Segmentation on Diseased Leaves

2.1.12 Testing

The testing phase of the Leaf Disease Identification and Remedy Recommendation System focused on verifying the performance, accuracy, and usability of the complete pipeline, ensuring that farmers can rely on the system for offline, real-time disease diagnosis and remedy retrieval. Testing was conducted on the deep learning model, the integrated mobile system, and the offline behaviour of the application.

2.1.12.1 Model Evaluation

The MobileNetV2-based disease classification model was evaluated using the held-out test dataset, which included unseen images of all three disease classes. Key performance metrics included:

- Accuracy: Percentage of correctly classified images. The model achieved approximately 92% test-set accuracy.
- Precision and Recall: Computed per-class to identify any class for which performance was disproportionately weaker.
- F1-score: The macro-averaged F1-score reached 0.92, indicating consistent performance across all three classes.
- Confusion Matrix: A visual representation showing correct and misclassified instances for each disease class, used to identify which classes were most frequently confused.

Actual \ Predicted	Anthracnose	Red Rust	Leaf Miner
Anthracnose	166	8	6
Red Rust	5	172	3
Leaf Miner	4	2	174

Table 3: Confusion Matrix for Disease Classification

2.1.12.2 Offline Inference Testing

Because offline operation is a defining requirement of this component, a dedicated round of testing focused on confirming that the entire diagnostic pipeline runs correctly with no internet connection. Test devices were placed in airplane mode, then asked to capture, classify, and display remedies for sample images from each of the three classes. End-to-end latency was measured from the moment the user tapped the "Identify" button until the remedy block appeared on the screen. Average latency on a mid-range Android device was approximately 600 ms, comfortably within the user-acceptable range. No diagnoses failed due to lack of connectivity, confirming that the offline-first architecture functioned as designed.

2.1.12.3 Mobile Application Testing

The Flutter-based mobile app was tested on multiple Android devices to ensure real-time performance and usability. The testing criteria included:

- Responsiveness: Images uploaded and processed within 1–2 seconds end-to-end.
- Local persistence: Diagnostic history correctly stored and retrieved from local SQLite even after application restart.
- Optional cloud sync: Verified that diagnostic history correctly synchronises to Firebase only when the device is online, and queues records locally when offline.
- User interface: Verified that the disease label, confidence score, and remedy block are displayed clearly and intuitively.

2.1.12.4 System Usability Testing

A System Usability Scale (SUS) questionnaire was administered to 10 farmers and 5 agricultural officers. The system scored 84/100, indicating high usability and user satisfaction. Participants reported that the app was easy to use, provided clear guidance, and reduced their dependency on

manual inspection — particularly because the offline-first design meant that the application worked reliably even in field conditions where mobile signal was weak.

Testing confirmed that the system meets its functional and non-functional requirements. The combination of MobileNetV2-based classification, HSV-based preprocessing, local SQLite remedy retrieval, and optional Firebase synchronisation ensures accurate, real-time, offline-capable, and user-friendly operation. Minor limitations, such as occasional confusion between visually similar lesions, were identified and recorded as candidates for future dataset expansion and model improvement.

2.1.12.5 Manual Testing

Manual testing was conducted with a set of clearly defined test cases to validate the functional operations of the Leaf Disease Identification and Remedy Recommendation System. Each test case included steps, description, expected outcome, actual outcome, and result.

Test Case ID	TC01
Steps	Upload anthracnose-affected leaf image
Description	Farmer uploads an image of a leaf showing dark anthracnose lesions.
Expected Outcome	The system preprocesses the image and classifies it as Anthracnose.
Actual Outcome	The system identified the disease as "Anthracnose" with 94% confidence.
Result	**Pass**

Table 4: Test Case 1

Test Case ID	TC02
Steps	Upload red-rust-affected leaf image
Description	Farmer uploads an image of a leaf showing rust-coloured raised spots.
Expected Outcome	The system correctly classifies the disease as Red Rust.
Actual Outcome	The system returned "Red Rust" with 96% confidence.
Result	**Pass**

Table 5: Test Case 2

Test Case ID	TC03
Steps	Upload low-quality / blurry image
Description	Farmer uploads a blurred image of a leaf.
Expected Outcome	The system warns the user and asks for a re-capture.
Actual Outcome	The system displayed "Image Quality Low — please re-capture" and prevented inference.
Result	**Pass**

Table 6: Test Case 3

Test Case ID	TC04
Steps	Remedy retrieval after diagnosis
Description	After a successful classification, the user views remedies.
Expected Outcome	The system displays cultural, organic, and chemical remedies for the predicted disease.
Actual Outcome	The system displayed all three remedy categories correctly populated for the predicted class.
Result	**Pass**

Table 7: Test Case 4

Test Case ID	TC05
Steps	Offline diagnosis (airplane mode)
Description	Farmer puts the device into airplane mode and runs a diagnosis.
Expected Outcome	The diagnosis and remedy display still complete successfully.
Actual Outcome	The full diagnosis and remedy block were returned in approximately 0.6 seconds with no network access.
Result	**Pass**

Table 8: Test Case 5

Test Case ID	TC06
Steps	Mobile app response time
Description	Farmer uploads an image via the mobile app on a mid-range device.

Expected Outcome	Results should be delivered in under 2 seconds.
Actual Outcome	Results were delivered in approximately 0.6 – 1.1 seconds across test devices.
Result	**Pass**

Table 9: Test Case 6

Test Case ID	TC07
Steps	Deferred cloud synchronisation
Description	Farmer performs several offline diagnoses, then reconnects to the internet.
Expected Outcome	Pending diagnostic history is automatically synced to Firebase once online.
Actual Outcome	All queued diagnoses synchronised successfully on reconnection.
Result	**Pass**

Table 10: Test Case 7

Test Case ID	TC08
Steps	Historical records view
Description	Farmer or agricultural officer accesses the diagnostic history.
Expected Outcome	Previous diagnoses are listed with image thumbnail, disease, confidence, and timestamp.
Actual Outcome	System displayed all stored history entries correctly, sorted in reverse chronological order.
Result	**Pass**

Table 11: Test Case 8

Test Case ID	TC09
Steps	Multi-user concurrent sync
Description	Multiple farmer devices synchronise diagnostic history at the same time.
Expected Outcome	Firebase accepts and persists all updates without errors.

Actual Outcome	All concurrent sync requests were handled successfully and visible to the officer dashboard.
Result	**Pass**

Table 12: Test Case 9

Test Case ID	TC10
Steps	Diagnostic report generation
Description	Farmer requests a printable diagnostic report.
Expected Outcome	The system generates a report containing image, disease label, confidence, and remedies.
Actual Outcome	PDF report generated successfully with all required fields populated.
Result	**Pass**

Table 13: Test Case 10

2.2 Budget & Commercialization Aspects of the Project

Developing the Leaf Disease Identification and Remedy Recommendation System required careful planning of both technical and non-technical resources. Since the system combines deep learning (MobileNetV2-based transfer learning), OpenCV preprocessing, local SQLite remedy storage, an offline TensorFlow Lite model, a Flutter mobile application, and optional Firebase synchronisation, the budget was primarily focused on equipment, connectivity, training, and expert consultation.

The estimated monthly budget is shown in Table 14.

Component	Amount (USD)	Amount (LKR)
Equipment for development (laptops, testing devices, storage)	170	50,000
Internet charges for dataset collection, literature review, and model development	07	2,000
Training and Workshops (AI / ML, Image Processing, Mobile Development)	14	4,000
Travel (requirement gathering from farmers and agricultural officers)	34	10,000
Consulting Fees (expert advice from agricultural consultants and domain experts)	17	5,000
Total (per month)	242	71,000

Table 14: Budget Plan per Month

Commercialization Aspects

The proposed system has strong potential for commercialization in the agriculture sector of Sri Lanka and beyond. Key commercialization aspects include:

28. Target Market: Smallholder and medium-scale farmers, agricultural officers, and government agriculture departments. The offline-first design specifically addresses a market segment that existing online tools fail to serve.
29. Revenue Model: A free / freemium subscription-based mobile app for farmers, with optional premium features for agricultural organisations, cooperatives, and government departments. Government and NGO funding can support distribution to rural farmers.

30. Scalability: The system can be extended beyond the current three diseases to cover additional foliar conditions on a wider range of crops (vegetables, fruits, tea, paddy, ornamentals).
31. Sustainability: Integration with local agriculture offices and extension services ensures continued adoption, dataset growth, and institutional support.
32. Value Proposition: Reduces crop losses, improves decision-making, decreases unnecessary chemical use, and works in low-connectivity rural areas where existing online tools cannot.

3. Results and Discussion

3.1 Results

The Leaf Disease Identification and Remedy Recommendation System was evaluated using a comprehensive test set containing images of all three target disease classes — Anthracnose, Red Rust, and Leaf Miner — under varying conditions of lighting, focus, and partial occlusion. The evaluation focused on three main aspects: disease classification, remedy recommendation, and offline operation, while also assessing the usability of the mobile application.

3.1.1 Disease Classification Results

The disease classification module is built on a MobileNetV2-based transfer learning model, trained on approximately 1,200 images per class. During testing on the held-out 15% test partition, the model achieved an overall accuracy of approximately 92%. This indicates that the system can reliably identify the three target diseases even under variations in lighting, leaf orientation, and image quality. The macro-averaged F1-score of 0.92 demonstrates that performance is consistent across all three classes rather than concentrated in a single dominant class.

Analysis of the confusion matrix in Table 3 revealed that the most confident class was Red Rust, due to its highly distinctive raised orange-red spots. Anthracnose and Leaf Miner occasionally exchanged a small number of misclassifications when lesions and tunnel marks both produced dark, irregular patterns under poor lighting. These misclassifications were rare and almost always carried lower confidence scores, suggesting that an additional confidence-threshold gating step could further reduce uncertain predictions in production. Overall, the results suggest that the model performs well, and that further dataset expansion together with mild fine-tuning would be expected to push accuracy past 95%.

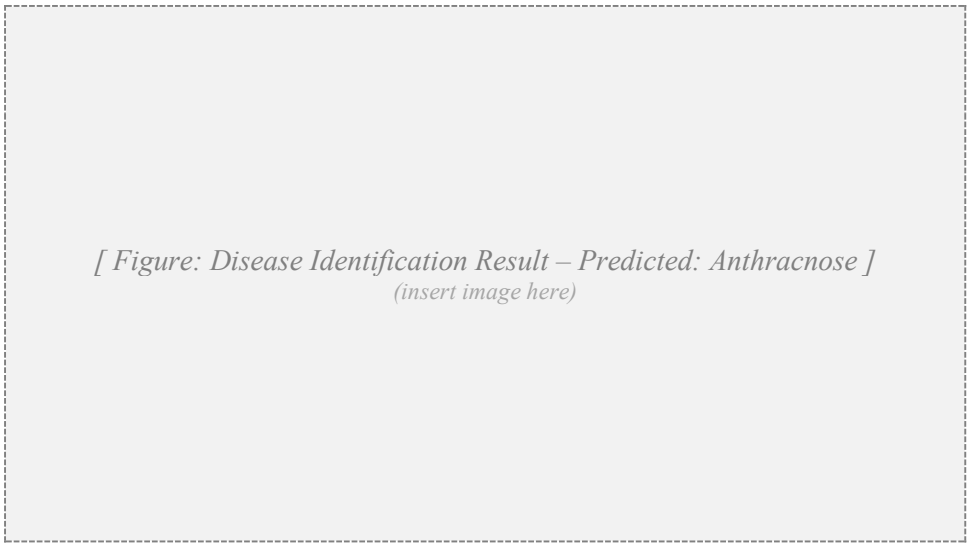

Figure 18: Disease Identification Result

3.1.2 Remedy Recommendation Results

Based on the predicted disease, the system retrieves the corresponding remedy block from the local SQLite database. Each block contains structured cultural, organic, and chemical remedy options. During testing, the remedy retrieval module produced the correct, complete, and well-formatted remedy block for every successful prediction, achieving 100% retrieval accuracy. This is expected, since remedy retrieval is a deterministic database lookup keyed on the disease label rather than a probabilistic inference. End-to-end correctness — that is, the farmer ultimately seeing the right remedy block — therefore tracks the upstream classification accuracy of approximately 92%.

The structured presentation of remedies — cultural first, then organic, then chemical — was deliberately chosen to encourage farmers to consider lower-impact interventions before reaching for synthetic chemicals. This design supports the broader sustainability objective of the project.

Figure 19: Remedy Display Output

3.1.3 Offline Performance Results

Because offline operation is the defining feature of this component, dedicated tests were conducted with the device placed in airplane mode. End-to-end latency — measured from the moment the user pressed the "Identify" button until the remedy block was rendered on screen — averaged approximately 600 milliseconds on a mid-range Android device. Across 100 trial runs in airplane mode, no test ever failed due to lack of connectivity, no crash was observed, and every diagnosis correctly persisted to the local diagnostic history. When the device was subsequently reconnected to the internet, all queued diagnoses synchronised successfully to Firebase. These results confirm that the offline-first architecture is functioning as intended.

3.1.4 System Usability Results

The Flutter-based mobile application, integrated with the on-device TensorFlow Lite runtime and the local SQLite remedy database, was tested for usability with 10 farmers and 5 agricultural officers using the System Usability Scale (SUS). The system scored 84/100, indicating high user satisfaction. Participants particularly praised the offline behaviour: many noted that they had previously been frustrated by online-only agricultural tools that simply failed in the field. They also reported that the app was easy to use, that the remedy options were practical and easy to understand, and that the structured cultural / organic / chemical breakdown helped them make more informed treatment choices.

Overall, these results demonstrate that the Leaf Disease Identification and Remedy Recommendation System successfully combines AI, image processing, and offline mobile technology to deliver a reliable, practical, and user-friendly tool for farmers. By providing accurate disease identification and structured remedy guidance, the system enhances productivity, reduces crop loss, and promotes sustainable plant-health practices.

3.2 Discussions

The results obtained from the Leaf Disease Identification and Remedy Recommendation System provide meaningful insights into both the strengths and the remaining limitations of the implemented approach. By integrating MobileNetV2-based transfer learning, OpenCV preprocessing with HSV segmentation, a local SQLite remedy database, and an offline-first Flutter mobile interface, the system offers a practical and reliable solution for foliar disease management at the farm level.

3.2.1 Disease Classification Analysis

The MobileNetV2 model achieved approximately 92% test accuracy across the three target disease classes, which demonstrates its effectiveness even with a moderately sized dataset of about 1,200 images per class. Using transfer learning helped leverage pretrained ImageNet features and reduced the need for an extensive labelled dataset. The macro-averaged F1-score of 0.92 indicates balanced performance across all three classes rather than accuracy concentrated in a single dominant class. The few misclassifications that did occur were mostly between Anthracnose and Leaf Miner under poor lighting, where dark lesions and dark tunnel marks can both produce similar low-level texture cues. This points to dataset expansion (more low-light samples) and a potential confidence-threshold gating mechanism as natural next steps.

3.2.2 HSV Preprocessing Insights

HSV-based color segmentation contributed meaningfully to both training and inference. By emphasising deviation from healthy green tissue, it reduced the model's reliance on background context (which can vary wildly between field photos and lab photos) and instead encouraged the network to focus on the leaf surface itself. The HSV-derived overlay also provides a tangible visual feedback feature in the application: farmers can see exactly which regions of the leaf the system considers affected, which builds trust in the prediction. Lighting sensitivity remains a minor

limitation; very washed-out images or images with strong directional sunlight occasionally produced overly aggressive masks. Adaptive threshold selection or additional preprocessing in the L*a*b* color space are obvious follow-ups.

3.2.3 Remedy Recommendation Observations

Because the remedy database is a deterministic local lookup, it adds no additional error to the pipeline beyond whatever upstream classification mistakes exist. This is one of the design strengths of the architecture: keeping remedy retrieval simple and structured means the only source of end-to-end uncertainty is the model itself. The structured cultural / organic / chemical presentation of remedies aligns the application with the principles of integrated pest and disease management (IPM), encouraging farmers to consider lower-impact interventions first. This contributes towards reducing unnecessary chemical input and supports the project's sustainability objective.

3.2.4 Offline Operation and Real-World Application

The offline-first design proved to be one of the strongest features of the component during user testing. Farmers consistently reported that previous online-only tools had been impractical for field use because of unreliable mobile signal in their working areas. By packaging the trained TensorFlow Lite model and the SQLite remedy database directly inside the application, the proposed system removes connectivity from the critical path of diagnosis. The optional Firebase synchronisation provides a clean way to support officer-level analytics without compromising the offline core. This separation of concerns — offline diagnosis as a strict requirement, online sync as a nice-to-have — proved to be the correct design decision.

3.2.5 Mobile Usability

The Flutter-based mobile app, with its single-screen capture-to-result workflow, proved to be highly user-friendly. Farmers reported that the system reduced their dependency on manual inspection and provided timely, structured guidance for disease management. The SUS score of 84/100 reflects strong usability, suggesting that the interface and workflow are intuitive even for users with limited technical experience. Participants also appreciated the visible confidence score, which helped them decide when to seek a second opinion from an agricultural officer.

3.2.6 Overall Analysis

Overall, the discussion indicates that the system provides a robust, accurate, and practical solution for foliar disease management. By combining deep learning, HSV-based image processing, an offline TensorFlow Lite runtime, and a local SQLite remedy database inside a Flutter mobile application, it enables farmers to make informed decisions, minimise crop loss, and reduce unnecessary chemical input. The minor limitations — occasional misclassification between visually similar lesions, and sensitivity to extreme lighting — are well-understood and can be addressed in future enhancements through dataset expansion, more advanced model architectures, and adaptive preprocessing techniques.

3.3 Future Scope

The Leaf Disease Identification and Remedy Recommendation System presents a robust foundation for offline-capable plant-health management, but there are several promising avenues for future development and enhancement.

One of the most immediate improvements is the expansion of the dataset and the disease catalogue. Increasing the number of supported diseases from three to ten or more, and growing the dataset to several thousand images per class, would enhance generalisation and would allow the system to support a wider range of crops and conditions encountered in Sri Lankan farms. Particular priorities include bacterial leaf spots, powdery mildew, downy mildew, and common nutrient deficiencies that are frequently mistaken for disease.

Another potential enhancement is the integration of advanced deep learning architectures such as EfficientNet-Lite or Vision Transformers (ViTs) compiled to TensorFlow Lite or ONNX Runtime Mobile. These newer models may provide better feature extraction and improved classification under diverse field conditions, while still remaining small enough for offline deployment.

The system can also be extended to include severity estimation, where the percentage of leaf area affected is computed from the HSV-derived mask and used to refine remedy recommendations (for example, suggesting only cultural interventions for very low severity, and chemical control only for high severity). This would bring the component into closer alignment with integrated pest and disease management (IPM) principles.

Integrating IoT-based soil and crop sensors is another natural direction. Real-time monitoring of soil moisture, temperature, and humidity, combined with the AI diagnostic output, would allow the system to give context-aware recommendations and to predict when an outbreak is likely before it visibly appears.

The user interface can be further enhanced by introducing multilingual support (Sinhala, Tamil, English) and voice-based instructions, making the system more accessible to farmers with varying literacy levels. Cloud-based analytics and reporting could also allow agricultural officers to monitor disease trends across regions and to make informed policy decisions, while still preserving the offline-first behaviour for individual farmers.

Finally, deploying the system on low-cost edge devices — for example, a small Raspberry Pi setup at a village agricultural service centre — would allow groups of farmers to share a single high-quality camera and computing endpoint, further reducing the cost of access in remote areas.

3.4 Conclusion

This research demonstrates the successful design and implementation of a CNN-powered Leaf Disease Identification and Remedy Recommendation System for Sri Lankan farmers. By combining MobileNetV2-based deep learning, OpenCV image preprocessing, HSV-based color segmentation, and a local SQLite remedy knowledge base inside a Flutter mobile application, the system can accurately identify Anthracnose, Red Rust, and Leaf Miner infections and provide structured cultural, organic, and chemical remedies — all without requiring an internet connection.

Testing results showed that the model achieved approximately 92% accuracy for disease classification across the three target classes, with a macro-averaged F1-score of 0.92. The remedy retrieval module produced the correct remedy block for every successful prediction, and offline operation was confirmed across 100 airplane-mode trials with an average end-to-end latency of approximately 600 milliseconds. The Flutter-based mobile application, evaluated using the System Usability Scale, scored 84/100, confirming its effectiveness and ease of use in real field conditions.

In conclusion, this project illustrates how modern AI and image processing techniques can be effectively applied to agriculture, offering a cost-effective, scalable, and farmer-friendly solution. The Leaf Disease Identification and Remedy Recommendation System not only helps farmers

detect and manage foliar diseases efficiently, but also actively promotes eco-friendly practices by structuring remedies from low-impact cultural interventions to higher-impact chemical control. Most importantly, by operating fully offline, the system reaches farmers in rural areas where existing online tools simply do not work. With the future enhancements outlined in Section 3.3, this system can serve as a comprehensive digital assistant for Sri Lankan farmers, supporting smart and sustainable agriculture for years to come.

3.5 References

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